

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-01-033

WASTE DISCHARGE REQUIREMENTS
FOR
AMADOR COUNTY PUBLIC WORKS AGENCY
AMADOR WATER AGENCY
AMADOR COUNTY SERVICE AREA NO.3
LAKE CAMANCHE VILLAGE WASTEWATER TREATMENT PLANT
AMADOR COUNTY

The California Regional Water Quality Control Board, Central Valley Region (hereafter Board), finds that:

1. The Amador County Public Works Agency submitted a Report of Waste Discharge (RWD), dated 1 September 2000, for updating Waste Discharge Requirements for the Amador County Service Area No. 3, Lake Camanche Village Wastewater Treatment Plant. Supplemental information was received on 9 November 2000 and 18 January 2001.
2. The existing collection system, wastewater treatment facility, and effluent disposal field (on Assessor's Parcel Number 003-042-097) are owned by the Amador County Public Works Agency and operated by the Amador Water Agency. Both agencies are hereafter known as Discharger.
3. Lake Camanche Village is situated approximately eight miles south of the City of Ione on the north side of Lake Camanche. The wastewater treatment plant is located 1/2 a mile north of Lake Camanche Village on County Road No. 88. The treatment plant is located in Section 26, T 5N, R9E, MDM&M, as shown on Attachment A, which is attached hereto and made part of the Order by reference.
4. Order No. 87-075, adopted by the Board on 24 April 1987, prescribes requirements for the Lake Camanche Village Wastewater Treatment Plant. This Order is neither adequate nor consistent with the current plans and policies of the Board.
5. The wastewater treatment plant receives domestic wastewater from approximately 243 residential units. Lake Camanche Village is currently approaching 60% of complete build out of the 395 parcels available. However, some remaining undeveloped parcels may be further subdivided and are zoned for multi-residential units and light commercial. The RWD predicts the growth rate for the community to be around 3.5% for the next ten years. A formal master plan for the community sewer system has never been developed.
6. The collection system has been built for the entire development with one gravity lateral connection for each parcel. The major components of the collection system include four lift stations and four force mains.
7. The treatment plant consists of a 7 acre-foot aerated facultative pond, two 0.08 acre-foot settling basins, chlorination facilities including a 240 foot long concrete lined chlorine

contact channel, a 19.5 acre-foot unlined storage reservoir, and a 12 acre effluent spray disposal system, as shown on Attachment B which is attached hereto and made part of the Order by reference.

8. The previous Order required effluent monitoring on a monthly basis for BOD₅ and total settleable solids. Based on the Discharger's self monitoring data, the average characteristics of the treated wastewater is as follows:

<u>Constituent</u>	<u>Units</u>	<u>Concentration</u>
BOD ₅	mg/l	17.6
Total Settleable Solids	ml/l•hr	<0.1

9. The current wastewater inflow to the treatment plant is unknown. The September 2000 RWD reports an average monthly dry weather flow of 23,355 gallons per day (gpd), an average monthly wet weather flow of 80,680 gpd, a peak daily dry weather flow of 97,200 gpd, and a peak daily wet weather flow of 120,200 gpd. However, the November 2000 addendum to the RWD presents a calculated dry weather flow of 97,200 gpd. During a December 2000 meeting, the Discharger stated that its flow measurements are inaccurate and that the actual dry weather flow is estimated to be 45,000 gpd. Recent self-monitoring reports state that the inflow fluctuates between 114,000 gpd and 30,000 gpd. Because the actual inflow is unknown, this Order does not contain a flow limitation.
10. The aerated facultative pond was designed for complete build out of the subdivision with a design flow of 281,250 gallons per day. However, the sprayfield and storage reservoir were not designed for build out. The Discharger submitted a preliminary water balance with the RWD. The calculations show that the storage reservoir does not have enough capacity to contain all flows from seasonal precipitation using a 100-year return period. The preliminary water balance did not include capacity calculations for increased flows due to growth.
11. The RWD states that wastewater overflows from the collection system and/or the storage reservoir occurred during 1995, 1998, 1999, and 2000 in violation of the discharge prohibition. The Amador County Grand Jury also reported that overflows from the storage reservoir occurred in 1997.
12. The Discharger has a history of sand-bagging the storage reservoir spillway in order to increase the amount of storage capacity in an effort to prevent additional spills. While this measure is acceptable in an emergency situation, it is not acceptable as a routine practice each winter. In addition, the Discharger installed a "sand filter" during the winter of 1997 to filter the wastewater before it was discharged to surface waters. This practice is not authorized by this Order.

13. The Discharger plans to add additional storage for treated wastewater immediately upstream of the existing reservoir by creating a new storage reservoir. The Discharger also plans to add additional land for sprayfield disposal. The Discharger has been trying for several years to acquire an additional 17.95 acres for effluent disposal from the adjacent property owner. However, the RWD reports that a timeline for construction is unknown at this time due to the stage of negotiations. Because the Discharger has not yet complied with the California Environmental Quality Act for these changes, the proposed expansion is not authorized under this Order.
14. The Discharger disposes of effluent through a sprinkler system on the 12 acre sprayfield. The sprinkler system consists of 66 sprinkler heads with an application rate of 300 gallons per hour per sprinkler head. The sprinklers are controlled with an automatic timer which is set to run from 8 to 12 hours per day depending on field conditions. The sprayfield vegetation consists of natural grass and bushes. No crops have ever been planted in the sprayfield.
15. The Discharger plans to install an additional two acre sprayfield that will be located immediately south of the existing sprayfield. The new sprayfield shall consist of 19 sprinkler heads, a tailwater recovery ditch along the southern portion of the sprayfield, and a portable pump for returning tailwater to the storage reservoir. The property is owned by Wayne and Shirley Shotwell and is currently used for cattle grazing. The Discharger has satisfied the California Environmental Quality Act by filing a categorical exemption for use of this land as sprayfield.
16. As described in the RWD, the application rates for the sprayfield are:

<u>Constituent</u>	<u>Loading Rate</u>
Flow	5.25 acre-feet/acre/year
Total Dissolved Solids	8,486 lb/acre/year
Total Nitrogen	77 lb/acre/year
BOD	220 lb/acre/year

17. Tailwater runoff from disposal area is captured and returned to the storage reservoir. The storage reservoir has two surface drains, the North and South diversion ditches, which traverse around the sides of the reservoir. The South diversion ditch is equipped with two valves for directing flow. When the sprayfield is irrigated, the valves are opened and tailwater from the south side of the sprayfield is directed back to the storage reservoir. When the Discharger ceases irrigation during rain events, the valves are closed and stormwater runoff from the sprayfield is diverted around the storage reservoir and discharged to an adjacent surface watercourse. However, there are no valves on the North diversion ditch. Therefore, excessive irrigation of the north side of the sprayfield can lead to discharges of wastewater to surface waters via the North ditch.

18. An inspection on 11 August 2000 found that the North diversion ditch was discharging wastewater from irrigation runoff to the surface watercourse. Staff observed that the sprayfield soil was saturated and a significant volume of tailwater was still running off the spray field 15 hours after irrigation had ceased. A Notice of Violation was issued, requiring the Discharger to cease discharging to the surface watercourse.
19. Due to the over application of wastewater on the sprayfield, and the potential to transport pollutants offsite, the Monitoring and Reporting Program requires the Discharger to collect surface water samples from the diversion ditches.
20. The Discharger disinfects the wastewater applied to the sprayfield using chlorination; however, the previous Order did not contain effluent limitations to ensure adequate disinfection. This Order contains an effluent limitation for total coliform organisms.
21. Staff inspections have found excessive weed growth and cattails around the treatment plant, storage reservoir, and diversion ditches. This Order requires the Discharger to establish a weed minimization program.
22. The collection system has a history of sewage overflows. A report by the Amador County Environmental Health Department states that all four lift stations overflowed during a power failure in 1995. However, the Discharger has only installed power receptacles for backup power supply in two of the four lift stations.
23. An inspection on 11 August 2000 found that the lift stations did not have adequate safeguards to prevent sewage overflows, had a record of poor maintenance, and that the Discharger only inspected the lift stations five days a week. Staff recommended that the Discharger improve the lift stations by installing remote communication alarm systems. However, the Discharger did not upgrade the lift stations, and on 4 September 2000, found that sewage was discharging from Lift Station C to an adjacent watercourse. Based on staff's interpretation of the Discharger's flow records, the lift station spilled for several days before a downstream property owner noticed the problem.
24. On 12 September 2000, the Board issued a Notice of Violation (NOV) for the sewage overflow. The NOV required that the Discharger improve the safeguards at Lift Station C. The Discharger responded by installing a light and horn, and posting a sign with the telephone number to call when the light was active. However, this action is inadequate to prevent overflows.
25. On 13 October 2000, the Executive Officer issued Administrative Civil Liability Complaint No. 5-00-526 to the Discharger for repeated sewage overflows from lift stations operated by the County of Amador, including lift stations at Lake Camanche Village.

26. The collection system experiences significant amounts of infiltration during the wet season. During an inspection, staff found that pipe inserts into manholes and wet wells have not been properly sealed, and observed manholes set at grade level in grassy swales and other drainage areas. The RWD states that that obvious infiltration problems will be corrected when discovered and that video inspection and smoke testing are expected to be part of a long range plan for detecting infiltration problems. The RWD states that the Discharger's five year plan for collection system improvements is only to upgrade Lift Station C.
27. The 1999-2000 Amador County Grand Jury report found that in the county service areas "it is evident that little or no routine, or preventative maintenance is performed. This conclusion is reinforced by the lack of a written preventative maintenance plan." In particular, the Grand Jury found that the Lake Camanche Village wastewater service area contained the following deficiencies:
- *"None of the four sewage lift stations has security fencing.*
 - *Lift station "C" is located adjacent to a creek which flows directly to Lake Camache. Power failures have resulted in overflows in the past.*
 - *A review of sewage flow charts at the wastewater treatment plant indicates the probability of high rates of infiltration and inflows. Unconfirmed reports indicate an average dry weather flow of 20 to 30 gallons per minute (gpm). Average wet weather flows can be as high as 50 gpm, indicating an infiltration problem. Peak instantaneous winter flows can be as high as 400 gpm, indicative of major sources of direct inflow.*
 - *In April, 1999, a survey was made by the California Rural Water Association of the extent of sludge deposits in the aerated lagoon at the wastewater treatment plant. Their report indicated a significant build up of sludge with an accompanying reduction in the capacity of the pond. They also noted symptoms of anaerobic conditions developing in the pond.*
 - *The effluent pond reportedly overflowed in 1997 and 1998. In the winter, ground is saturated, flows to the pond are high and cloud cover reduces evaporation; all factors which lead to the possibility of future overflows."*
28. The Discharger has not removed sludge from the aerated facultative pond in over twenty-five years. As a result, the sludge has accumulated on the bottom of the pond to a depth of approximately five feet (about 40% of the pond's total storage capacity). The sludge is impacting pond capacity and may affect effluent quality. During an inspection on 11 August 2000, staff observed surfacing sludge in the aerated facultative pond and around the effluent overflow weir. In late summer 2000, the Discharger proposed to remove the

sludge and bury it onsite. However, the Discharger did not provide any data showing that this disposal method would not impact water quality.

29. The RWD states that sludge will be removed from the pond. The Discharger proposes to pump the sludge out of the pond, dewater it, and bury it in a shallow pit onsite. This proposal would create a sewage sludge disposal unit. The information provided with RWD was insufficient to allow the Discharger to proceed with the project.
30. The Discharger does not receive trucked or hauled waste, or waste from recreational vehicle dump stations.
31. The treatment plant and disposal area is within a 1,000 feet of a public drinking water well, Amador County Well No. 12. In addition, a private drinking water well that serves two residential units on the adjacent property is within 500 feet of the disposal area. The treatment plant and sprayfield is within 2,500 feet of Lake Camanche, a recreational area.
32. Federal regulations for stormwater discharges were promulgated by the U.S. Environmental Protection Agency on 16 November 1990 (40 CFR Parts 122, 123, and 124). The regulations require specific categories of facilities which discharge stormwater associated with industrial activities to obtain NPDES permits. The flow at this wastewater treatment plant is less than 1.0 mgd and therefore the Discharger is not required to apply for a stormwater NPDES permit.
33. The Basin Plan encourages reclamation.
34. Surrounding land uses are zoned for agriculture and residential. The residential housing density is zoned for five acre parcels.
35. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento River and San Joaquin River Basins (hereafter Basin Plan), which contains beneficial uses, water quality objectives for waters of the Basins, and implementation plans. These requirements implement the Basin Plan.
36. Surface water drainage is to Lake Camanche.
37. The beneficial uses of Lake Camanche are municipal and domestic supply; agricultural supply; contact and other noncontact recreation; warm and cold freshwater habitat; migration for warm water species; spawning for warm and cold water species; and wildlife habitat.
38. The beneficial uses of underlying groundwaters are municipal, industrial, and agricultural supply.

39. The Board has considered anti-degradation pursuant to State Water Resources Control Board Resolution No. 68-16 and finds that not enough data exists to determine whether this discharge is consistent with those provisions. Therefore, this Order provides a timeline for data collection to determine whether the discharge will cause an increase in groundwater or surface water constituents above that of background levels. If the discharge is causing such an increase, then the Discharger may be required to cease the discharge, line the ponds, implement source control, change the method of disposal, or take other action to prevent groundwater degradation.
40. The action to update waste discharge requirements for this facility is exempt from the provisions of the California Environmental Quality Act (CEQA), in accordance with Title 14, California Code of Regulations (CCR), Section 15301.
41. This discharge is exempt from the requirements of *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 2005, et seq., (hereafter Title 27). The exemption pursuant to Section 20090(b), is based on the following:
 - a. The Board is issuing waste discharge requirements,
 - b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to Title 22 CCR, Division 4.5, and Chapter 11, as a hazardous waste.
42. Section 13267(b) of California Water Code provides that: "In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of discharging, or who proposes to discharge within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste outside of its region that could affect the quality of the waters of the state within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the board requires. The burden, including costs of these reports, shall bear a reasonable relationship to the need for the reports and the benefits to be obtained from the reports."
43. The Board has consulted with the State Department of Health Services and the Amador County Environmental Health Department and considered their recommendations regarding public health aspects for the use of reclaimed water.
44. The Board has notified the Discharger, and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.

45. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 87-075 is rescinded and the Amador County Public Works Agency, Amador Water Agency, Amador County Service Area No. 3, Lake Camanche Village Wastewater Treatment Plant, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

Note: Other prohibitions, conditions, definitions, and methods of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated 1 March 1991.

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Neither the treatment nor the discharge shall cause a nuisance or condition of pollution as defined by the California Water Code, Section 13050.
4. The discharge shall not cause the degradation of any water supply.
5. Discharge of waste classified as hazardous, as defined in Sections 2521(a) of Title 23, CCR, Section 2510, et seq., (hereafter Chapter 15), or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.
6. Surfacing of wastewater outside or downgradient of the ponds is prohibited.
7. Excessive irrigation with reclaimed water, or continued irrigation of reclaimed water during periods of precipitation, is prohibited.
8. Discharge of untreated or partially treated waste to groundwater is prohibited.
9. The discharge of any wastewater other than that from domestic sources or domestic equivalent is prohibited.
10. Application of wastewater to areas different than those described in Findings No. 14 and 15 is prohibited.

11. The discharge of trucked or hauled wastes including waste from recreational vehicles into the system is prohibited.
12. The use of the new two acre sprayfield is prohibited unless the Discharger has both a tailwater recovery ditch and pump in place.

B. Discharge Specifications:

1. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.
2. As a means of discerning compliance with Discharge Specification No. 1, the dissolved oxygen content in the upper zone (1 foot) of all wastewater in ponds shall not be less than 1.0 mg/l.
3. The wastewater treatment and storage ponds shall not have a pH of less than 6.5 or greater than 8.5.
4. The Discharger shall operate all systems and equipment to maximize treatment of wastewater and optimize the quality of the discharge.
5. The ponds shall be managed to prevent the breeding of mosquitoes. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the waste surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, and/or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
6. The Discharger's wastewater treatment system shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
7. The freeboard in the storage reservoir shall never be less than two feet as measured vertically from the water surface to the lowest point of overflow.
8. The wastewater ponds shall have sufficient capacity to accommodate allowable wastewater flow, design seasonal precipitation, and ancillary inflow and infiltration. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with the historical rainfall patterns.

9. On or about 1 October each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specifications No. 7 and No. 8.

C. Effluent Limitations

The discharge of effluent to the disposal field in excess of the following limits is prohibited:

Parameter	30-day Avg. Limit	Daily Maximum Limit
BOD ₅ ¹	40 mg/L	80 mg/L
Total Suspended Solids	60 mg/L	
Total Nitrate (as Nitrogen)	10 mg/L	
Total Coliform Organisms	23 MPN/100 ml	230 MPN/100 ml
Settleable Solids	0.5 ml/l	

¹ 5-day 20° C Biochemical Oxygen Demand

D. Reclamation Requirements:

1. In all unsecured areas, reclaimed wastewater conveyance lines shall be clearly marked as such. Reclaimed water controllers, valves, etc. shall be affixed with reclaimed water warning signs, and these and quick couplers and sprinkler heads shall be of a type, or secured in such a manner, that permits operation by authorized personnel only.
2. Public contact with reclaimed wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.
3. Areas irrigated with reclaimed water shall be managed to prevent breeding of mosquitoes. More specifically:
 - a. All applied irrigation water must infiltrate completely within 48-hours.
 - b. Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.
 - c. Low-pressure and un-pressurized pipelines and ditches which are accessible to mosquitoes shall not be used to store reclaimed water.
4. Reclaimed water used for irrigation shall be managed to minimize both erosion and runoff from the disposal area.
5. Direct or windblown spray shall be confined to the designated reclamation area and prevented from contacting drinking water facilities.

6. The Discharger may not irrigate with effluent 24 hours before precipitation, during periods of precipitation, and for at least 24 hours after cessation of precipitation, or when soils are saturated.
7. The Discharger may not spray irrigate with effluent when wind velocities exceed 30 mph.
8. To alert the public of the use of reclaimed water, signs with proper wording of sufficient size shall be placed at areas of access and around the perimeter of all areas used for effluent disposal.
9. There shall be no further impoundment of reclaimed water within 500 feet of any domestic water well or irrigation well unless it is demonstrated to the satisfaction of the Executive Officer that a shorter distance is justified.
10. No spray irrigation of any recycled water, other than disinfected tertiary recycled water, shall take place within 100 feet of a residence or place where public exposure could be similar to that of a park, playground, or school yard.

E. Solids Disposal Requirements:

1. Collected screenings, grit, sludge, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer, and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq.
2. Storage, use and disposal of sewage sludge shall comply with existing Federal, State, and local laws and regulations, including permitting requirements and technical standards included in 40 CFR Part 503 and the Statewide General Order for the Discharge of Biosolids (Water Quality Order No. 2000-10-DWQ) (or any subsequent document which replaces Order No. 2000-10-DWQ).
3. Sludge and other solids shall be removed from ponds, clarifiers, etc. as needed to ensure optimal plant operation and adequate hydraulic capacity. Drying operations shall take place such that leachate does not impact groundwater or surface water.
4. If biosolids will be stored onsite between 15 October and 15 May of any year, then they shall be stored in a facility constructed in accordance with Class II surface impoundment or waste pile standards contained in Title 27 of the CCR, or similar facility approved by the Executive Officer. Such a facility shall be designed and maintained to prevent inundation or washout from a storm or flood with a 100 year return frequency. The Discharger shall collect any leachate or stormwater that comes in contact with the biosolids pile and return it to the wastewater treatment plant.

5. Biosolids may not be disposed of onsite except as a soil amendment used at agronomic rates. Before onsite disposal is permitted, the Discharger must submit a technical report containing all the information required for a Notice of Intent for Water Quality Order No. 2000-10-DWQ (or any subsequent document which replaces Order No. 2000-10-DWQ). The technical report must be submitted at least 60 days before the anticipated date of disposal.
6. Disposal of biosolids at a permitted municipal solid waste landfill or at a permitted publicly owned treatment works is acceptable. The Discharger may also elect to dispose of its biosolids at a facility permitted under Order No. 2000-10-DWQ or at a similar facility permitted under individual WDRs. No matter where the biosolids are taken, the Discharger must comply with all sampling and analytical requirements of the entity that accepts the waste.

F. Groundwater Limitations:

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentration statistically greater than background water quality, except for coliform bacteria. For coliform bacteria, increases shall not cause the most probable number of total coliform organisms to exceed 2.2 MPN/100 ml over any 7-day period.

G. Surface Water Limitations:

The discharge shall not cause the surface watercourse downstream of the storage reservoir to contain waste constituents in concentrations statistically greater than background (i.e., upstream) surface water quality.

H. Provisions:

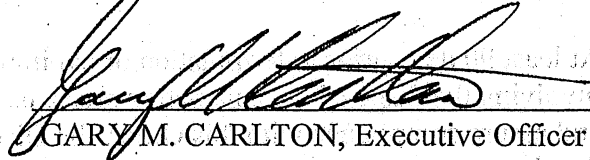
1. All of the following reports shall be submitted pursuant to Section 13267 of the California Water Code.
 - a. By **31 March 2001**, the Discharger shall submit a weed minimization control plan for controlling weeds and cattails around the treatment plant and storage reservoir.
 - b. By **30 June 2002**, the Discharger shall submit a Groundwater Monitoring Well Workplan. The workplan shall be prepared by a Registered Geologist and shall contain the information listed in Attachment C: *Items to be Included in a Monitoring Well Installation Workplan and a Monitoring Well Installation Report of Results.*

- c. By **31 August 2001**, the Discharger shall submit a report showing that it has installed a valve or similar control structure on the North diversion ditch in order to prevent discharges of irrigation tailwater to surface watercourses.
 - d. By **30 June 2003**, the Discharger shall submit a Monitoring Well Installation Report prepared by a Registered Geologist. The report shall describe the installation of groundwater monitoring wells and shall contain the items found in the second section of Attachment C.
2. The Discharger shall comply with the Monitoring and Reporting Program No. 5-01-033.
 3. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
 4. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving the disposal or reclamation areas, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
 5. The Discharger shall submit to the Board on or before each compliance report due date the specified document, or if appropriate, a written report detailing compliance or noncompliance with the specific schedule date and task. If noncompliance is reported, then the Discharger shall state the reasons for noncompliance and shall provide a schedule to come into compliance.
 6. The Discharger shall use the best practicable cost-effective control technique(s) currently available to comply with discharge limits specified in this order.
 7. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge.
 8. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, then the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to this office.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-01-033
AMADOR COUNTY PUBLIC WORKS AGENCY
AMADOR WATER AGENCY
AMADOR COUNTY SERVICE AREA NO. 3
LAKE CAMANCHE VILLAGE WWTP
AMADOR COUNTY

9. The Discharger shall comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
10. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
11. The Board will review this Order periodically and may revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 26 January 2001.


GARY M. CARLTON, Executive Officer

Attachments

ASB:26 January 2001

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 5-01-033
FOR
AMADOR COUNTY PUBLIC WORKS AGENCY
AMADOR WATER AGENCY
AMADOR COUNTY SERVICE AREA NO. 3
LAKE CAMANCHE VILLAGE WASTEWATER TREATMENT PLANT
AMADOR COUNTY

This Monitoring and Reporting Program (MRP) describes requirements for monitoring domestic wastewater, groundwater, surface water, and biosolids. This MRP is issued pursuant to Water Code Section 13267. The Discharger shall not implement any changes to this MRP unless and until a revised MRP is issued by the Executive Officer. Specific sample station locations shall be approved by Regional Board staff prior to implementation of sampling activities.

All samples should be representative of the volume and nature of the discharge or matrix of material sampled. The time, date, and location of each grab sample shall be recorded on the sample chain of custody form.

INFLUENT MONITORING

Samples shall be collected at the same frequency and at approximately the same time as effluent samples and should be representative of the influent at the headworks prior to treatment. Influent monitoring shall include, at a minimum, the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	gallons	Continuous	Daily	Monthly
BOD ₅ ¹	mg/l	Grab	Quarterly	Quarterly
Total Suspended Solids	mg/l	Grab	Quarterly	Quarterly

¹ BOD₅ denotes five-day, 20° Celsius Biochemical Oxygen Demand.

EFFLUENT MONITORING

Effluent samples shall be collected downstream from the last connection through which wastes can be admitted to the storage pond. Grab samples are considered adequately composited to represent the effluent. Effluent monitoring shall include, at a minimum, the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Total Chlorine Residual	mg/l	Grab	Daily	Monthly
BOD ₅ ¹	mg/l	Grab	Weekly	Monthly
Total Settleable Solids	ml/l•hr	Grab	Weekly	Monthly
Total Coliform Organisms ²	MPN/100 ml	Grab	Weekly	Monthly
Total Suspended Solids	mg/l	Grab	Weekly	Monthly
Total Dissolved Solids	mg/l	Grab	Monthly	Monthly
Total Kjeldahl Nitrogen	mg/l	Grab	Monthly	Monthly
Nitrates as Nitrogen	mg/l	Grab	Monthly	Monthly

¹ BOD₅ denotes five-day, 20° Celsius Biochemical Oxygen Demand.

² Using a minimum of 15 tubes or three dilutions

SPRAYFIELD MONITORING

Inspections of the spray disposal area shall be conducted daily (during operation) and the results shall be included in the monthly monitoring report. Evidence of erosion, field saturation, runoff, or the presence of nuisance conditions shall be noted in the report. Reclaimed water shall also be monitored to ascertain loading rates at the disposal area. Monitoring of the disposal fields shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	gallons	Continuous	Daily	Monthly
Rainfall	Inches	Measurement	Daily	Monthly
Acreage Applied ¹	Acres	Calculated	Daily	Monthly
Application Rate	gal/acre/day	Calculated	Daily	Monthly
Total Nitrogen	lbs/month	Calculated	Monthly	Monthly
Total Dissolved Solids	lbs/month	Calculated	Monthly	Monthly

¹ Sprayfields shall be identified

POND MONITORING

Samples shall be collected from an established sampling station located in an area that will provide a sample representative of the wastewater in the aerated pond, settling ponds, and storage reservoir. Freeboard will be measured vertically from the surface of the storage reservoir to the lowest elevation of the surrounding berm and shall be measured to the nearest 0.25 feet. Monitoring of all four ponds shall include, at a minimum, the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Freeboard ¹	Feet	Measurement	Semi-weekly ²	Monthly
Dissolved Oxygen	mg/l	Grab	Weekly	Monthly
PH	pH units	Grab	Weekly	Monthly

¹Storage reservoir only.

²Semi-weekly shall mean twice per week.

GROUNDWATER MONITORING

Prior to construction of any groundwater monitoring wells, the Discharger shall submit plans and specifications to the Board for review and approval. Once installed, all new wells shall be added to the MRP, and shall be sampled and analyzed according to the schedule below.

Prior to sampling groundwater elevations shall be measured and the wells shall be purged at least three well volumes until pH and electrical conductivity have stabilized. Depth to groundwater shall be measured to the nearest 0.01 feet. Water table elevations shall be calculated and used to determine groundwater gradient and direction of flow. Samples shall be collected using approved EPA methods. Groundwater monitoring shall include, at a minimum, the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling and Reporting Frequency</u>
Groundwater elevation	Feet	Measurement	Quarterly
pH	S.U.	Grab	Quarterly
Total Dissolved Solids	mg/l	Grab	Quarterly
Total Kjeldahl Nitrogen	mg/l	Grab	Quarterly
Nitrates as Nitrogen	mg/l	Grab	Quarterly
Ammonia as Nitrogen	mg/l	Grab	Quarterly
Total Coliform Organisms ¹	MPN/100 ml	Grab	Quarterly

¹ Using a minimum of 15 tubes or three dilutions

² Beginning with the third quarter 2003.

SURFACE WATER MONITORING

The following surface water monitoring program shall apply only when visible water is present in the surface watercourse or either diversion ditch. As shown on Attachment B, the Discharger shall establish four sampling stations: one station (S-1) shall be 100 feet upstream of the easternmost storage reservoir and spray field, one station (S-2) shall be at the North diversion ditch just prior to its discharge into the surface watercourse, one station (S-3) shall be at the South diversion ditch just prior to its discharge into the surface watercourse, and one station (S-4) shall be on the surface watercourse at the downstream property boundary. Samples of the surface water shall be analyzed for the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling and Reporting Frequency</u>
Total Dissolved Solids	mg/l	Grab	Monthly
Total Coliform Organisms ¹	MPN/100 ml	Grab	Monthly
Total Fecal Organisms ¹	MPN/100 ml	Grab	Monthly

¹ Using a minimum of 15 tubes or three dilutions

BIOSOLIDS MONITORING

The Discharger shall keep records regarding the quantity of biosolids generated by the treatment processes; any sampling and analytical data; the quantity of biosolids stored on site; and the quantity removed for disposal. The records shall also indicate that steps taken to reduce odor and other nuisance conditions. Records shall be stored onsite and available for review during inspections.

If biosolids are transported off-site for disposal, then the Discharger shall submit records identifying the hauling company, the amount of biosolids transported, the date removed from the facility, the location of disposal, and copies of all analytical data required by the entity accepting the waste. If biosolids are disposed of onsite, then the Discharger shall submit the annual report information as contained in the Statewide General Order for the Discharge of Biosolids (Water Quality Order No. 2000-10-DWQ) (or any subsequent document which replaces Order No. 2000-10-DWQ).

All records shall be submitted as part of the Annual Monitoring Report.

COLLECTION SYSTEM AND PUMP STATION MONITORING

The Discharger shall monitor the collection system and pump stations on its property on a monthly basis for pipe anomalies, cracks, overflows, or leaks. In addition, the Discharger shall also report on

the operational status of the pump station pumps, alarm systems, and any emergency or backup equipment that is used to prevent the loss of control of waste from the collection system. This information shall be submitted in the monthly reports.

REPORTING

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., effluent, pond, etc.), and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Regional Board.

As required by the California Business and Professions Code Sections 6735, 7835, and 7835.1, all Groundwater Monitoring Reports shall be prepared under the direct supervision of a Registered Engineer or Geologist and signed by the registered professional.

A. Monthly Monitoring Reports

Monthly reports shall be submitted to the Regional Board on the **1st day of the second month following sampling** (i.e. the January Report is due by 1 March). At a minimum, the reports shall include:

1. Results of influent, effluent, pond, sprayfield, collection system, and surface water monitoring.
2. A comparison of monitoring data to the discharge specifications and an explanation of any violation of those requirements. Data shall be presented in tabular format.
3. A statement describing conditions around the treatment ponds and storage reservoir relative to the amount of vegetation.
4. If requested by staff copies of laboratory analytical report(s).

B. Quarterly Monitoring Reports

The Discharger shall establish a quarterly sampling schedule for groundwater and influent monitoring such that samples are obtained approximately every three months. Quarterly monitoring reports shall be submitted to the Board by the **1st day of the second month after the quarter** (i.e. the January-March quarter is due by May 1st) each year. The Quarterly Report shall include the following:

1. Results of groundwater and influent monitoring. The results of regular monthly monitoring reports for March, June, September and December may be incorporated into their corresponding quarterly monitoring report.

2. A narrative description of all preparatory, monitoring, sampling, and analytical testing activities. The narrative shall be sufficiently detailed to verify compliance with the WDR, this MRP, and the Standard Provisions and Reporting Requirements. The narrative shall be supported by field logs for each well documenting depth to groundwater; parameters measured before, during, and after purging; method of purging; calculation of casing volume; and total volume of water purged;
3. Calculation of groundwater elevations, an assessment of groundwater flow direction and gradient on the date of measurement, comparison of previous flow direction and gradient data, and discussion of seasonal trends if any;
4. A narrative discussion of the analytical results for all media and locations monitored including spatial and temporal trends, with reference to summary data tables, graphs, and appended analytical reports (as applicable);
5. A comparison of monitoring data to the groundwater limitations and an explanation of any violation of those requirements;
6. Summary data tables of historical and current water table elevations and analytical results;
7. A scaled map showing relevant structures and features of the facility, the locations of monitoring wells and any other sampling stations, and groundwater elevation contours referenced to mean sea level datum;
8. Copies of laboratory analytical report(s) for groundwater monitoring.

C. Annual Report

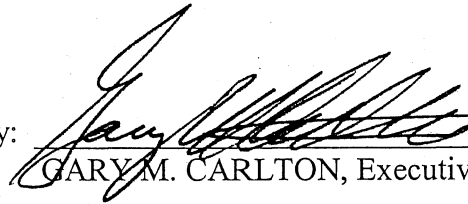
An Annual Report shall be prepared as the fourth quarter monitoring report. The Annual Report will include all monitoring data required in the monthly/quarterly schedule. The Annual Report shall be submitted to the Regional Board by **1 February** each year. In addition to the data normally presented, the Annual Report shall include the following:

1. The contents of the regular quarterly monitoring report for the last quarter of the year.
2. If requested by staff, tabular and graphical summaries of all data collected during the year;
3. An evaluation of the performance of the domestic wastewater treatment system, as well as a forecast of the flows anticipated in the next year;
4. An evaluation of the groundwater quality beneath the wastewater treatment facility.
5. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements.

6. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.
7. Summary of information on the disposal of biosolids as described in the "Biosolids Monitoring" section.
8. If biosolids were disposed onsite, then the annual report information as contained in the Statewide General Order for the Discharge of Biosolids (Water Quality Order No. 2000-10-DWQ) (or any subsequent document which replaces Order No. 2000-10-DWQ);
9. Whether the Discharger anticipates removing biosolids in the coming year, and if so, the anticipated schedule for cleaning, drying, and disposal;

The Discharger shall implement the above monitoring program as of the date of this Order.

Ordered by:


GARY M. CARLTON, Executive Officer

26 January 2001

(Date)

THE UNIVERSITY OF CHICAGO
DIVISION OF THE PHYSICAL SCIENCES
DEPARTMENT OF CHEMISTRY
530 SOUTH EAST ASIAN AVENUE
CHICAGO, ILLINOIS 60607
TEL: 773-936-5000

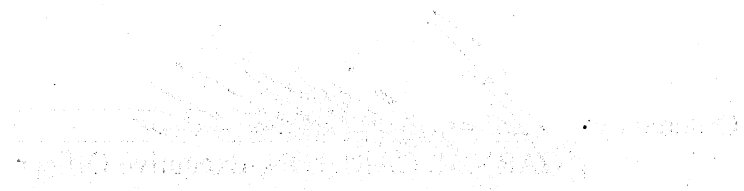
Dear Sirs:

I am pleased to inform you that your application for admission to the M.S. program in Chemistry has been accepted.

You will receive a letter from the Graduate School regarding the application process and the requirements for admission to the M.S. program.

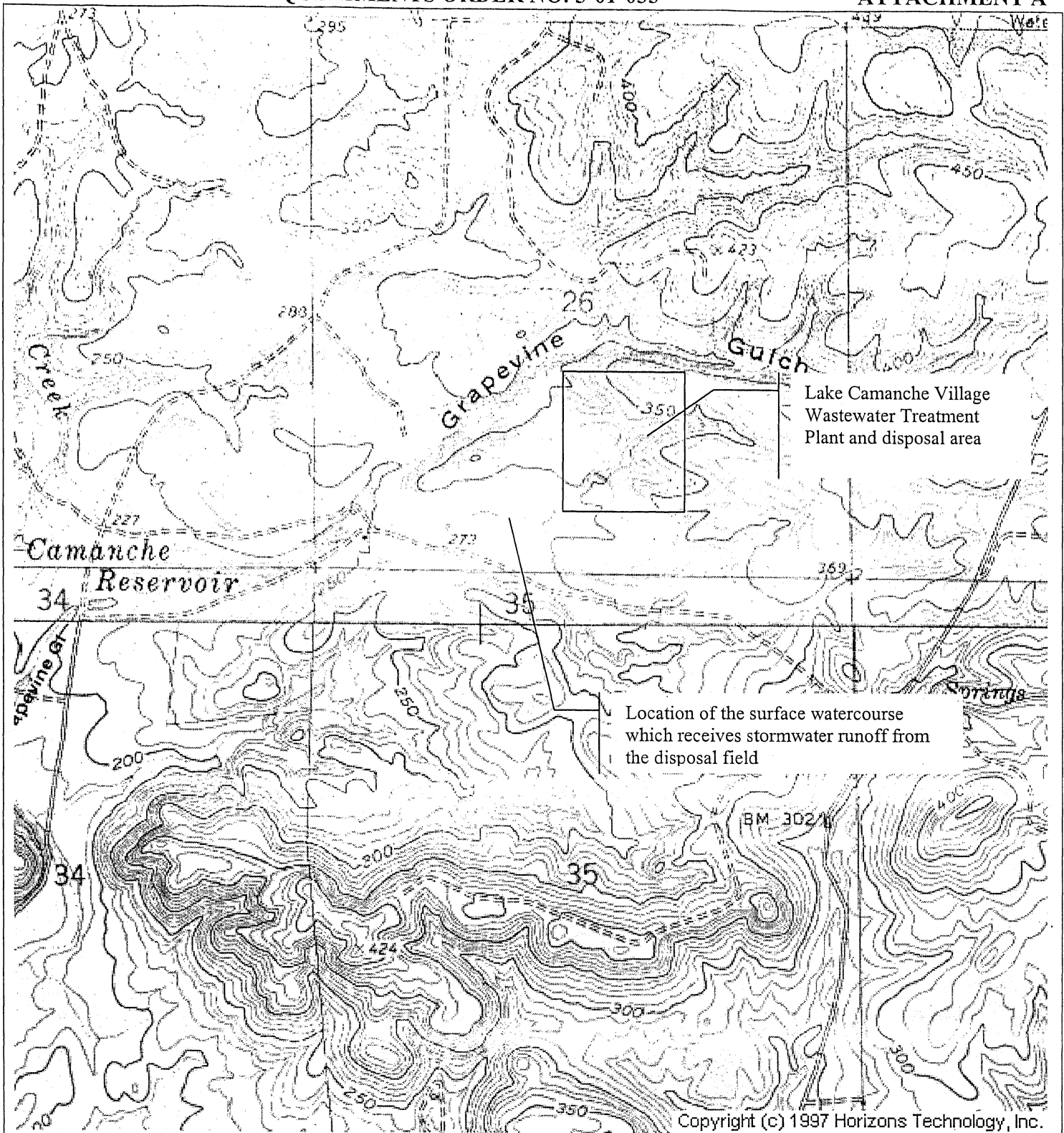
Please contact the Graduate School at 773-936-5000 for more information.

Sincerely,
The Graduate School



Very truly yours,
The Graduate School

cc: 100



Drawing Reference:

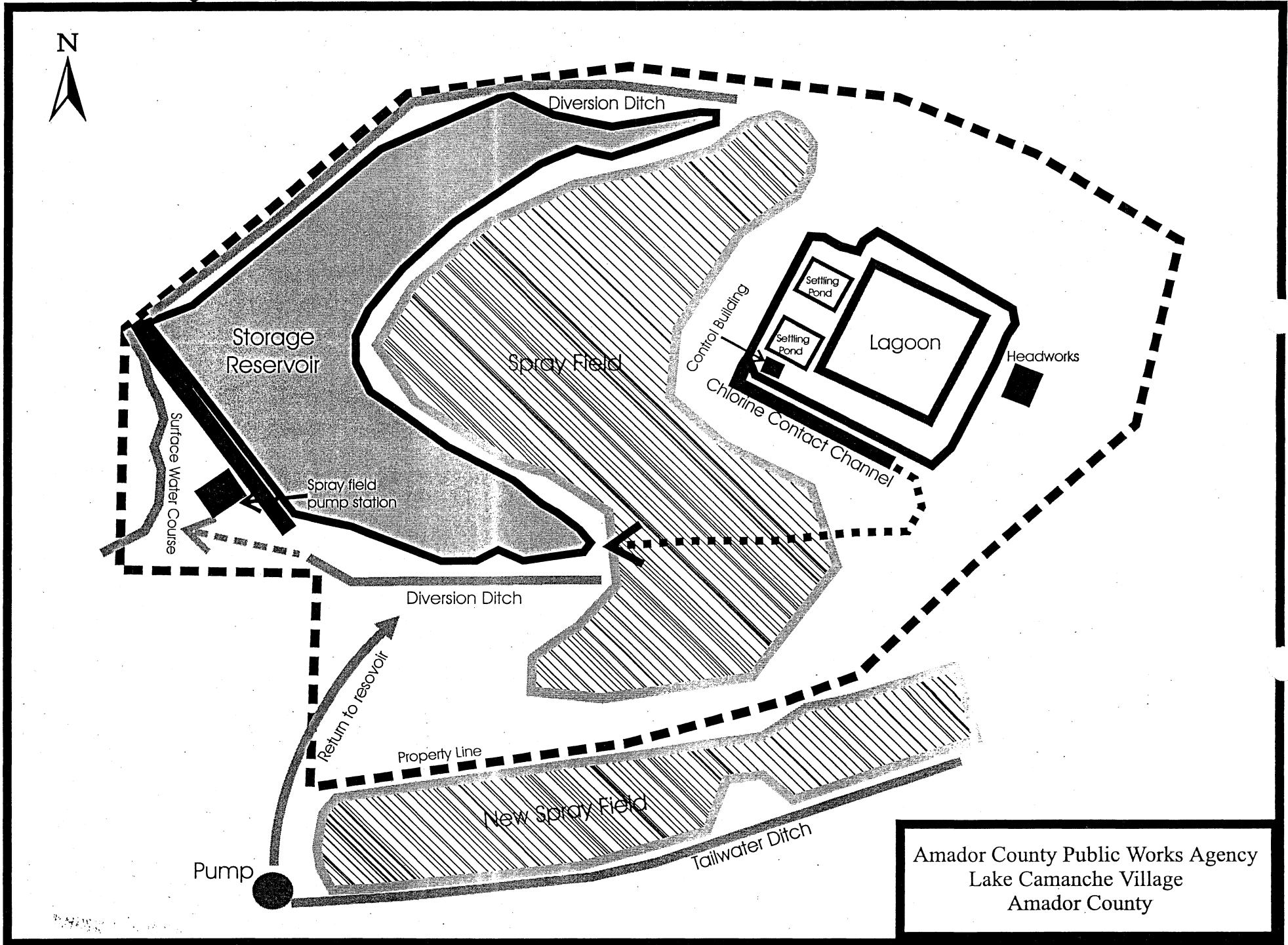
United States Department of the
Interior Geological Survey
Ione 7.5' Quadrangle
Section 26, T5N, R9E, MDB&B

SITE PLAN

AMADOR COUNTY PUBLIC WORKS
AGENCY, CSA #3
LAKE CAMANCHE VILLAGE WWTP
AMADOR COUNTY



1 in = 1333 ft



Amador County Public Works Agency
Lake Camanche Village
Amador County



NEW YORK PUBLIC LIBRARY ASTOR LENOX TILDEN FOUNDATION



Winston H. Hickox
Secretary for
Environmental
Protection

California Regional Water Quality Control Board

Central Valley Region

Robert Schneider, Chair



Gray Davis
Governor

Sacramento Main Office

Internet Address: <http://www.swrcb.ca.gov/rwqcb5>
3443 Rortier Road, Suite A, Sacramento, California 95827-3003
Phone (916) 255-3000 • FAX (916) 255-3015

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-01-033 ATTACHMENT C

ITEMS TO BE INCLUDED IN A MONITORING WELL INSTALLATION WORKPLAN AND A MONITORING WELL INSTALLATION REPORT OF RESULTS

Prior to installation of groundwater monitoring wells, the Discharger shall submit a workplan containing the minimum listed information. Wells may be installed after staff approve the workplan. Upon installation of the monitoring wells, the Discharger shall submit a report of results, as described below. All workplans and reports must be signed by a registered geologist, or certified engineering geologist.

Monitoring Well Installation Workplan

A. General Information:

- Monitoring well locations and rationale
- Survey details
- Equipment decontamination procedures
- Health and safety plan
- Topographic map showing any existing monitoring wells, proposed wells, waste handling facilities, utilities, and other major physical and man-made features.

B. Drilling Details: describe drilling and logging methods

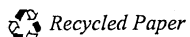
C. Monitoring Well Design:

- Casing diameter
- Borehole diameter
- Depth of surface seal
- Well construction materials
- Diagram of well construction
- Type of well cap
- Size of perforations and rationale
- Grain size of sand pack and rationale
- Thickness and position of bentonite seal and sand pack
- Depth of well, length and position of perforated interval

D. Well Development:

- Method of development to be used
- Method of determining when development is complete
- Method of development water disposal

California Environmental Protection Agency



- E. Surveying Details: discuss how each well will be surveyed to a common reference point
- F. Soil Sampling (if applicable):
 - Cuttings disposal method
 - Analyses to be run and methods
 - Sample collection and preservation method
 - Intervals at which soil samples are to be collected
 - Number of soil samples to be analyzed and rationale
 - Location of soil samples and rationale
 - QA/QC procedures
- G. Well Sampling:
 - Minimum time after development before sampling (48 hours)
 - Well purging method and amount of purge water
 - Sample collection and preservation method
 - QA/QC procedures
- H. Water Level Measurement:
 - The elevation reference point at each monitoring well shall be within 0.01 foot. Ground surface elevation at each monitoring well shall be within 0.1 foot. Method and time of water level measurement shall be specified.
- I. Proposed time schedule for work.

Monitoring Well Installation Report of Results

- A. Well Construction:
 - Number and depth of wells drilled
 - Date(s) wells drilled
 - Description of drilling and construction
 - Approximate locations relative to facility site(s)
 - A well construction diagram for each well must be included in the report, and should contain the following details:
 - Total depth drilled
 - Depth of open hole (same as total depth drilled if no caving occurs)
 - Footage of hole collapsed
 - Length of slotted casing installed
 - Depth of bottom of casing
 - Depth to top of sand pack
 - Thickness of sand pack
 - Depth to top of bentonite seal
 - Thickness of bentonite seal
 - Thickness of concrete grout
 - Boring diameter
 - Casing diameter
 - Casing material

- Size of perforations
- Number of bags of sand
- Well elevation at top of casing
- Depth to ground water
- Date of water level measurement
- Monitoring well number
- Date drilled
- Location

B. Well Development:

- Date(s) of development of each well
- Method of development
- Volume of water purged from well
- How well development completion was determined
- Method of effluent disposal
- Field notes from well development should be included in report.

C. Well Surveying: provide reference elevations for each well and surveyor's notes

D. Water Sampling:

- Date(s) of sampling
- How well was purged
- How many well volumes purged
- Levels of temperature, EC, and pH at stabilization
- Sample collection, handling, and preservation methods
- Sample identification
- Analytical methods used
- Laboratory analytical data sheets
- Water level elevation(s)
- Groundwater contour map

E. Soil Sampling (if applicable):

- Date(s) of sampling
- Sample collection, handling, and preservation method
- Sample identification
- Analytical methods used
- Laboratory analytical data sheets

1. *Chlorophyll a* (Chl *a*)
 2. *Chlorophyll b* (Chl *b*)
 3. *Chlorophyll c* (Chl *c*)
 4. *Chlorophyll d* (Chl *d*)
 5. *Chlorophyll e* (Chl *e*)
 6. *Chlorophyll f* (Chl *f*)
 7. *Chlorophyll g* (Chl *g*)
 8. *Chlorophyll h* (Chl *h*)
 9. *Chlorophyll i* (Chl *i*)
 10. *Chlorophyll j* (Chl *j*)
 11. *Chlorophyll k* (Chl *k*)
 12. *Chlorophyll l* (Chl *l*)
 13. *Chlorophyll m* (Chl *m*)
 14. *Chlorophyll n* (Chl *n*)
 15. *Chlorophyll o* (Chl *o*)
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 17. *Chlorophyll q* (Chl *q*)
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 30. *Chlorophyll ad* (Chl *ad*)
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 132. *Chlorophyll acv* (Chl *acv*)
 133. *Chlorophyll acw* (Chl *acw*)
 134. *Chlor*

INFORMATION SHEET

ORDER NO. 5-01-033
AMADOR COUNTY PUBLIC WORKS AGENCY
AMADOR COUNTY SERVICE AREA NO. 3
LAKE CAMANCHE VILLAGE WWTP
AMADOR COUNTY

Lake Camanche Village is about eight miles south of the City of Ione on the north side of Lake Camanche Reservoir in Amador County. The residential community is currently approaching 60% of build out and a 3.5% growth rate is predicted over the next ten years.

The treatment plant consists of an aerated facultative pond, two settling basins, chlorination facilities including a chlorine contact channel, an unlined storage reservoir, and an effluent spray irrigation system. The Discharger has not removed sludge from the aerated facultative pond in 25 years. The sludge has accumulated to a depth of approximately five feet and is impacting the treatment capacity of the pond.

The storage reservoir and disposal field are undersized. The water balance submitted by the Discharger shows that the storage reservoir has insufficient capacity for all flows from seasonal precipitation using a 100-year return period. The storage reservoir has a history of overflows. In the past, the Discharger has even sand bagged the spillway to increase storage capacity in an effort to prevent additional spills.

In addition, over-application of wastewater to the spray field has created saturated soil conditions and excessive tailwater runoff. The storage reservoir is equipped with two diversion ditches; however, only the South ditch has a valve for directing flow. When irrigating, the valve is opened and tailwater runoff is directed into the storage reservoir. During rain events, the valve is closed and stormwater runoff is allowed to flow past the storage reservoir and is discharged into the unnamed surface watercourse below the reservoir. However, the North ditch does not have a valve to direct flow, and excessive irrigation can lead to runoff entering the nearby surface watercourse. An August 2000 staff inspection found this type of discharge.

The collection system also has a history of sewage overflows and suffers from high infiltration flows during the wet season. Furthermore, staff inspections found that the lift stations lacked adequate safeguards to prevent spills and had a record of poor maintenance. On 13 October 2000, the Executive Officer issued Administrative Civil Liability Complaint No. 5-00-526 to the Discharger for repeated sewage overflows from lift stations operated by the County of Amador, including lift stations at Lake Camanche Village.

The Monitoring and Reporting Program (MRP) has been expanded from the previous Order to include surface water and groundwater monitoring. In order to comply with the MRP, the Discharger is required to install groundwater monitoring wells.

Due to the Dischargers inability to comply with the WDRs, Cease and Desist Order No. 5-01-034 was adopted concurrently with these WDRs.

